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A theory on thermal spalling of rocks with a focus on thermal spallation drilling

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Key Points:
- A model is presented, evaluating the required conditions to initiate spalling and characterizing rocks according to their ability to spall
- The model couples rapid heat transfer with induced thermal stresses and the fracture-based mechanics in rocks
- An experimental evaluation showed reasonable agreement to the values suggested by the model

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Abstract Thermal cracking of rocks is an intensively studied topic in different research areas and for various engineering problems. In this context, we present a modeling approach which describes thermal spalling of rocks with a focus on thermal spallation drilling. This drilling technology uses high thermal loads to locally destruct the surface of the rock formation. With the presented model, the operating conditions which are required to initiate spalling of rocks can be estimated. Additionally, a Spallability number is introduced allowing a categorization of rocks according to their ability to spall. The presented model is based on linear fracture mechanics and the stress intensity concept. It evaluates if a crack with a certain geometry embedded in a rock with specific properties propagates during exposure to different heat loads and external pressures. Thereby, rapid heat transfer processes are coupled with induced thermal stresses and fracture mechanics in rocks.

1. Introduction

Thermal crack creation and propagation of rocks is an intensively studied research area [*Heap et al.*, 2014; *Fredrich et al.*, 1986; *Lu and Fleck*, 1998; *Siddiqi and Evans*, 2015; *Wang et al.*, 1989]. One section of this research field focuses on the intended local disintegration of the rock surface, called spallation [*Williams et al.*, 1996; *Rauenzahn and Tester*, 1989; *Germanovich*, 1977]. This process is based on the effect that certain rocks will locally disintegrate into small fragments if rapidly exposed to high thermal loads. Thereby, due to thermally induced stresses, cracks initially present in the rock structure will propagate and connect with other cracks until a so-called spall is formed. Eventually, this spall will detach from the surface and expose the newly created surface [*Preston and White*, 1934; *Germanovich*, 1977; *Rauenzahn and Tester*, 1989] (see Figure 1). A possible application field of the spallation effect are thermal contactless drilling technologies [*Kant and Rudolf von Rohr*, 2016; *Hillson and Tester*, 2015; *Timoshkin et al.*, 2003; *Kocis et al.*, 2015; *Bybee*, 2006], where hot fluid jets, laser beams, or plasma jets are used as the required heat source. These technologies recently came into renewed research focus, as they have the potential to significantly reduce the drilling costs [*Tester et al.*, 2006; *Augustine*, 2009; *Kant et al.*, 2015] and many known problems have been overcome (e.g., ignition and drilling in an aqueous environment) [*Stathopoulos*, 2013; *Teodoriu and Cheuffa*, 2011; *Ndeda et al.*, 2015].

In order to optimize this drilling technology, the operating conditions which are required to initiate spalling of the present rock formation have to be known. Additionally, criteria are needed to classify rock types according to their ability to spall and to identify the application range where the process can be used. Different modeling approaches have been developed to describe these characteristics [*Rauenzahn and Tester*, 1989; *Germanovich*, 1977; *Hasselman*, 1969]. Nevertheless, none of these available models sufficiently consider the coupled process of thermal stress induction and fracture-based mechanics in rocks by additionally providing straightforward application possibilities without the need for additional calibration experiments. Therefore, we present a model based on linear fracture mechanics, the stress intensity concept, and the transient temperature distribution inside the rock structure. The discussed model can be used to evaluate the critical conditions required to initiate thermal rock fracturing and to map rocks according to their ability to spall. The criterion is based on the models of *Zhao et al.* [2000] and *Lu and Fleck* [1998], which have been adapted to spalling of rocks and extended to include influences due to prevailing confining pressures.

In this manuscript, first, the model principles, the mathematical formulation, and simplification possibilities are presented. Thereafter, the general characteristics of the model are evaluated with a sensitivity study and

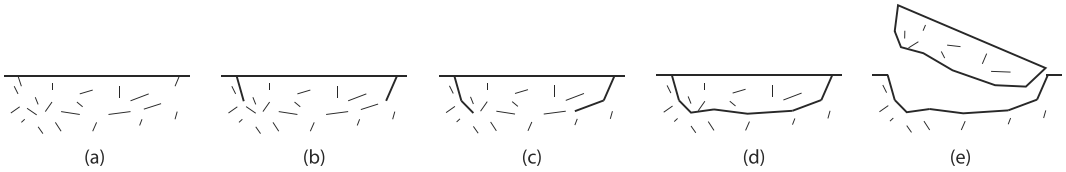


Figure 1. Principles of the spalling mechanism: (a) initial rock structure with microcracks, (b) crack propagation, (c) crack coalescence, (d) spall formation, and (e) spall ejection.

other parameters influencing spalling of rocks are discussed. As a last step, the proposed model is compared with spallation experiments conducted on a typical granitic rock. The applicability of this model is demonstrated, as the experimental surface temperature measurements for which spallation occurs are in reasonable agreement with the model.

2. Derivation of the Spalling Criterion

The presented spalling model is based on the verification, if the stress intensity factor at the tip of a surface crack exceeds the fracture toughness of the material. Thereby, the stress intensity factor is induced by thermal stresses due to the transient temperature distribution inside the body and mechanical stresses due to prevailing confining pressures.

The analyzed domain consists of a plane strain edge crack with the length a embedded in a half-space (see Figure 2). The surface of this half-space is exposed to a convective boundary condition with the heat transfer coefficient h and the fluid temperature T_{fl}. Additionally, a general constant pressure state described by a pressure component parallel (p) to the crack is homogeneously acting on the investigated crack.

The presented crack geometry was selected as it has a higher stress intensity factor than other crack geometries relevant for thermal spalling. Therefore, it propagates first at similar geometries and loading situations. A validation later discussed in the manuscript (section 4) shows the eligibility of this approach.

2.1. Transient Temperature Distribution

As a first step to derive the stress intensity factor at the tip of the presented crack, the transient temperature distribution inside the domain is discussed. The description of the temperature field in the presented half-space, due to a constant convective boundary condition, can be found in various literature sources [*Carslaw and Jaeger*, 1959; *Hahn and Oezisik*, 2012; *von Boeckh and Wetzel*, 2012] and is presented in equation (1),

$$\Theta = \Theta_{\mathrm{fl}} \left\{ \mathrm{erfc}\left[\frac{z}{\sqrt{4\kappa t}}\right] - \exp\left(\frac{hz}{\lambda} + \kappa t \frac{h^2}{\lambda^2}\right) \cdot \mathrm{erfc}\left(\frac{z}{\sqrt{4\kappa t}} + \frac{h}{\lambda}\sqrt{\kappa t}\right)\right\} \tag{1}$$

where $\Theta = (T - T_i)$ is the difference between the evaluated temperature and the initial temperature, $\Theta_{\mathrm{fl}} = (T_{\mathrm{fl}} - T_i)$ the difference between the fluid temperature and the initial temperature, z the distance from the surface,

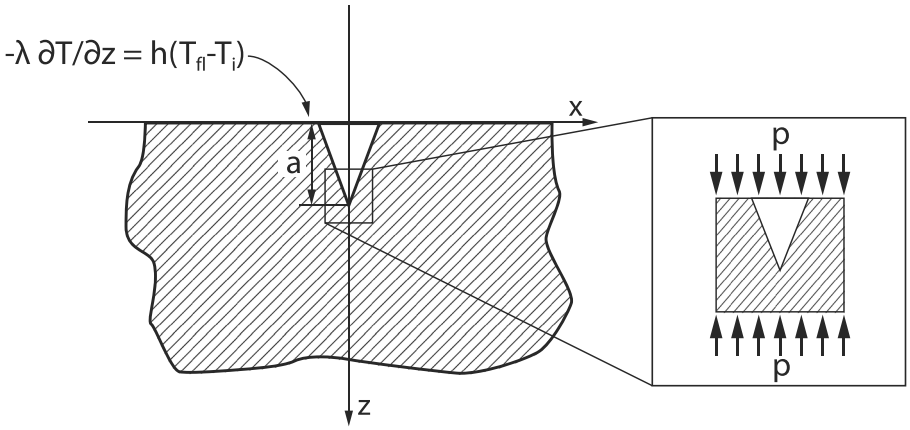


Figure 2. Half-space with an embedded crack, a convective boundary condition on the surface $z = 0$ and a vertical pressure state described by p.

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κ the thermal diffusivity, t the time, h the heat transfer coefficient, and λ the heat conductivity of the solid. Thereby, a potential influence of the crack on the temperature distribution is neglected and the boundary condition is assumed to be constant around the investigated crack. Additionally, the physical properties are assumed to be constant over the considered temperature range. In order to abbreviate this equation, several dimensionless variables are introduced: the dimensionless time τ, the Biot number Bi based on the crack length, and the dimensionless distance ξ (see equation (2)).

$$\tau = \frac{\kappa t}{a^2}, \quad Bi = \frac{h \cdot a}{\lambda}, \quad \xi = \frac{z}{a} \tag{2}$$

With these dimensionless numbers, the temperature distribution can be abbreviated to equation (3).

$$\Theta = \Theta_{\mathrm{fl}} \left\{ \mathrm{erfc}\left[\frac{\xi}{\sqrt{4\tau}}\right] - \exp\left(Bi\,\xi + \tau\, Bi^2\right) \cdot \mathrm{erfc}\left(\frac{\xi}{\sqrt{4\tau}} + Bi\sqrt{\tau}\right)\right\} \tag{3}$$

Concluding, the temperature distribution is a function of the fluid temperature and the introduced dimensionless factors and can be written as $\Theta = \Theta_{\mathrm{fl}} \cdot f_T(\tau, Bi, \xi)$.

2.2. General Stress State and Stress Intensity Concept

The general stress state around the crack is approximated with a plane strain consideration [*Hutchinson and Suo*, 1991; *Hetnarski et al.*, 2009] by following the principles of linear fracture mechanics. The general constitutive relation for this stress state is shown in equation (4) [*Hetnarski et al.*, 2009],

$$\begin{aligned}
\sigma_{xx} &= \frac{E}{(1-2\nu)(1+\nu)}\left((1-\nu)\epsilon_{xx} + \nu\epsilon_{zz}\right) - \frac{E\alpha\Theta}{(1-2\nu)} \\
\sigma_{zz} &= \frac{E}{(1-2\nu)(1+\nu)}\left((1-\nu)\epsilon_{zz} + \nu\epsilon_{xx}\right) - \frac{E\alpha\Theta}{(1-2\nu)} + p \\
\sigma_{xz} &= \frac{E}{1+\nu}\epsilon_{xz}
\end{aligned} \tag{4}$$

where E is the Young's modulus, ϵ_{ij} the individual strain components, σ_{ij} the individual stress components, ν the Poisson's ratio, α the coefficient of linear thermal expansion, p the pressure component (positive values point toward the investigated element, see Figure 2), and Θ again the difference between the evaluated temperature and the initial temperature. Thereby, it is assumed that the cold rock material around the crack acts as a confinement ($\epsilon_{xx} = \epsilon_{yy} = 0$) and on the surface is no additional stress than due to the vertical pressure component p ($\sigma_{zz} = 0$). The confinement condition can be ensured by sufficient large rock samples or by additional confinement pressure acting on the samples. These assumptions allow a simple stress consideration, with the resulting normal stresses in the x direction, which are responsible for the crack formation presented in equation (5).

$$\sigma_{xx} = -\frac{E\alpha}{(1-\nu)}\Theta - p\frac{\nu}{(1-\nu)} \tag{5}$$

The stress intensity factor acting on the tip of the presented plane strain edge crack of length a due to an inhomogeneous stress field can be calculated according to *Tada et al.* [2000] as shown in equation (6).

$$K_I = 2\sqrt{\frac{a}{\pi}}\int_0^1 -\sigma_{xx}\frac{1.3 - 0.3\xi^{5/4}}{\sqrt{1-\xi^2}}\mathrm{d}\xi = 2\sqrt{\frac{a}{\pi}}\int_0^1 -\sigma_{xx}\, f_K(\xi)\,\mathrm{d}\xi \tag{6}$$

With the formulation for the stress state inside the half-space (equation (5)) and the transient temperature distribution discussed in equation (3), the stress intensity factor at the crack tip can be further described as shown in equation (7).

$$K_I = 2\sqrt{\frac{a}{\pi}}\left(\frac{E\alpha\Theta_{\mathrm{fl}}}{(1-\nu)}\int_0^1 f_T(\tau, Bi, \xi)\cdot f_K(\xi)\,\mathrm{d}\xi + \int_0^1 \left(p\,\frac{\nu}{(1-\nu)}\right)\cdot f_K(\xi)\,\mathrm{d}\xi\right) \tag{7}$$

2.3. Definition of the Spallability Number

The crack turns unstable if the stress intensity factor exceeds the fracture toughness of the rock ($K_I \geq K_{IC}$) [*Irwin*, 1957; *Bieniawski*, 1967]. If this limit is exceeded, the crack will propagate through the solid toward the closest stress drain which could be another crack or the surface of the half-space. If the crack reaches the

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surface of the domain, a so-called spall is formed, which will lead to the characteristic disintegration of the surface [*Kant and Rudolf von Rohr*, 2016; *Preston and White*, 1934]. A dimensionless number called Spallability is introduced to describe this critical condition where crack formation is initiated. This Spallability number is defined as $Sp = K_I/K_{IC}$ and with the derived formulation for the stress intensity factor (equation (7)) the Spallability number can be written as shown in equation (8). If the critical value of $Sp = 1$ is reached or exceeded, the conditions for crack formation are fulfilled and the crack will propagate through the solid.

$$Sp = 2\sqrt{\frac{a}{\pi}}\left(\frac{E\alpha\Theta_{\mathrm{fl}}}{K_{IC}(1-\nu)}\int_0^1 f_T(\tau,Bi,\xi)\cdot f_K(\xi)\mathrm{d}\xi + \frac{1}{K_{IC}}\int_0^1\left(p\frac{\nu}{(1-\nu)}\right)\cdot f_K(\xi)\mathrm{d}\xi\right) \geq 1 \tag{8}$$

With all introduced abbreviations written explicitly, the Spallability number can be calculated according to equation (9):

$$Sp = 2\sqrt{\frac{a}{\pi}}\left(\frac{E\alpha\Theta_{\mathrm{fl}}}{K_{IC}(1-\nu)}\int_0^1\left\{\mathrm{erfc}\left[\frac{z}{\sqrt{4\kappa t}}\right] - \exp\left(\frac{hz}{\lambda}+\kappa t\frac{h^2}{\lambda^2}\right)\cdot\mathrm{erfc}\left(\frac{z}{\sqrt{4\kappa t}}+\frac{h}{\lambda}\sqrt{\kappa t}\right)\right\}\cdot \frac{1.3-0.3\xi^{5/4}}{\sqrt{1-\xi^2}}\mathrm{d}\xi + \frac{1}{K_{IC}}\int_0^1\left(p\frac{\nu}{(1-\nu)}\right)\cdot\frac{1.3-0.3\xi^{5/4}}{\sqrt{1-\xi^2}})\mathrm{d}\xi\right) \geq 1 \tag{9}$$

This equation can be used to assess the operating conditions (flame temperatures and heat transfer coefficients) required to initiate spalling of a certain rock formation and the time frame until the onset of spallation appears. The derived equations are only valid if the assumptions used to derive this model are not violated. The most important assumptions which have been used are (i) the physical properties can be evaluated at the temperature and stress state where the spall is formed and are constant along the crack length and around the crack, (ii) the stress state can be described by the presented simplified plane strain model, (iii) the temperature distribution is not significantly influenced by preexisting cracks, and (iv) the used formulation for the crack intensity factor describes sufficiently the fracture mechanism.

2.4. Simplifications for the Derived Spallability Number

The Spallability number defined in equation (8) contains several terms which are difficult to evaluate. Therefore, equation (8) will be simplified to enable convenient access to the derived Spallability criterion. The first integral can be accurately approximated (for $Bi < 10$) as shown in equation (10),

$$\int_0^1 f_T(\tau,Bi,\xi)\cdot f_K(\xi)\mathrm{d}\xi \approx \left(1-\frac{1}{1+Bi}\right)f_3(\tau) \tag{10}$$

where $f_3(\tau)$ is a term considering the transient change of the integral. As the Biot number is based on the crack length, this simplification does not imply a practical restriction. The time when the crack is initiated is irrelevant for the investigated problem. Therefore, the time dependence is neglected and the function $f_3(\tau)$ is simplified to a constant value representing the maximum value of this function for any dimensionless time τ (see equation (11) and Figure 3).

$$A^* = \max(f_3(\tau)), \forall\ \tau > 0 \tag{11}$$

For even lower Biot numbers (Bi < 0.5), equation (10) can be further simplified as shown in equation (12)

$$\int_0^1 f_T(\tau,Bi,\xi)\cdot f_K(\xi)\mathrm{d}\xi \approx A^*\cdot Bi \tag{12}$$

If the pressure component is assumed to be independent from the crack length, the second integral of equation (8) can be directly numerically calculated (see equation (13)).

$$\int_0^1 f_K\mathrm{d}\xi = \int_0^1\frac{1.3-0.3\xi^{5/4}}{\sqrt{1-\xi^2}}\mathrm{d}\xi = 1.763 \tag{13}$$

Therewith and with equation (12), the reduced Spallability number can be approximated as shown in equation (14).

$$Sp_{\mathrm{red}} = \underbrace{2A^*\sqrt{\frac{a^3}{\pi}}\frac{E\alpha}{K_{\mathrm{IC}}(1-\nu)\lambda}}_{Sp_{\mathrm{rock}}}\cdot\underbrace{(T_{\mathrm{fl}}-T_i)\cdot h}_{Sp_{\mathrm{fluid}}} + \underbrace{\frac{2\cdot 1.763}{K_{IC}}\sqrt{\frac{a}{\pi}}\left(p\frac{\nu}{(1-\nu)}\right)}_{Sp_{\mathrm{conf}}} \tag{14}$$

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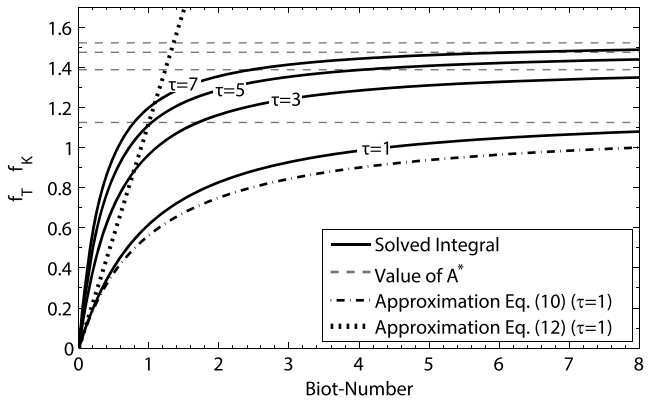


Figure 3. Comparison of the exact integral solution and the proposed approximation over a range of Biot numbers and dimensionless times.

The reduced formulation for the Spallability number of rocks can be divided into three parts. The first part (Sp_{rock}) is a function of rock properties, the second part (Sp_{fluid}) is only a function of heat transfer properties, and the last term (Sp_{conf}) is considering the influence of confining pressures on the crack tip. So in theory, rocks can be spalled, if the fluid properties are adjusted to account for the properties of the rock and the current pressure state. A low rock Spallability Sp_{rock} can be compensated with high fluid operating conditions Sp_{fluid}, in order to exceed the critical Spallability number of $Sp = 1$. Therewith, equation (14) can be used to map rocks according to their ability to spall. The only limit in this consideration is that the melting temperature should not be exceeded, as already partial melting will significantly inhibit thermal crack formation [*Rosenberg and Handy*, 2005; *Friedman et al.*, 1979].

3. Sensitivity Analysis and Characteristics of the Derived Spallation Criterion

A compact sensitivity analysis is performed, indicating the impact of the different involved properties on the discussed Spallability criterion and showing the general characteristics of the derived model. Thereby, the reduced Spallability number discussed in equation (14) is used, the value of every input property is consecutively increased to 10% and 20% from the reference value and the relative influence of this change on the resulting Spallability number is monitored. In order to normalize the influence, some properties were summarized to reference parameters (see Figure 4). The baseline values of all these reference parameters were set to one.

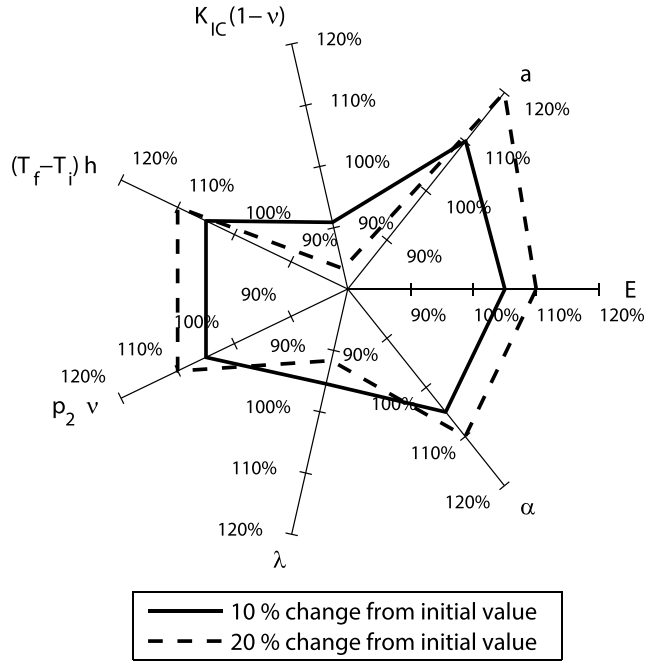


Figure 4. Sensitivity of the Spallability number on the required input parameters with $A^* = 1.763$; all other parameters are set to one.

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Figure 4 shows the result of this sensitivity study. It can be seen that several parameters (E, α, $(T_f - T_i) \cdot h$, $1/\lambda$) have the same linear influence on the result of the Spallability number. Nevertheless, as also equation (14) shows, the crack length has a considerably higher influence. Therefore, appropriate values of the initial present cracks in the rock are a crucial point for a precise calculation of the proposed Spallability number. Furthermore, increasing pressures parallel to the crack will increase the spallability and therefore significantly enhance the spallation process.

Additionally, the sensitivity analysis shows that the presented model follows and supports already observed and reported influences of the involved rock properties on the ability to spall. In order to initiate thermal spalling, the following rock properties are desired: low thermal conductivities [*Preston and White*, 1934; *Walsh and Lomov*, 2013], high Young's moduli (also described as hard rocks) [*Rauenzahn and Tester*, 1989; *Preston and White*, 1934; *Walsh et al.*, 2011], high vertical pressures [*Rauenzahn and Tester*, 1989; *Hoeser*, 2016], high thermal expansion coefficients [*Rauenzahn and Tester*, 1989; *Preston and White*, 1934], and low fracture toughness values (brittle rocks) [*Rauenzahn and Tester*, 1989; *Walsh et al.*, 2011].

Various other effects could potentially influence spalling and thermal cracking which are not considered in the derived spallation criterion. Significant increases of the thermal expansion during phase changes as, e.g., the α-β quartz transition appearing at 573°C [*Glover et al.*, 1995] will create additional mechanical stresses on preexisting cracks. If the affected minerals are confined by other rock material, the expansion will create further stresses on the rock structure which enhances spalling [*Van der Molen*, 1981].

On the other hand, several effects exist which can hinder the spallation effect. If the rock is heated above the brittle-to-ductile temperature, the rock will react with ductile deformation to the thermal stresses rather than spalling [*Byerlee*, 1968]. Additionally, if the rock is heated above the melting temperature, only insufficient thermal stresses can be created [*Rosenberg and Handy*, 2005; *Friedman et al.*, 1979]. Both effects originate from the effect that Young's modulus and fracture toughness will significantly decrease at the brittle-ductile transition or the melting temperature [*Heuze*, 1983]. Therefore, the lowest melting temperature of all minerals present in the rock structure determines the upper temperature limit where spallation can appear [*Kant and Rudolf von Rohr*, 2016; *Rauenzahn and Tester*, 1989]. Additionally, weak mineral components (as, e.g., in matrix supported rocks) or global fractures embedded in the rock structure will impede spalling of rocks, as the rock can expand into these components, preventing sufficiently high thermal stresses from being created.

4. Preliminary Validation With Central Aare Granite

In order to preliminarily validate the proposed spallability model, the temperature jump which the material has to be exposed to in order to initiate thermal crack formation was determined experimentally and compared with the values suggested by the model.

4.1. Experimental Procedure and Rock Material

The spallation experiments were conducted with the setup and measurement procedure described by *Kant and Rudolf von Rohr* [2016]. In this setup rock samples are rapidly exposed to a methane-air flame jet with a temperature of about 1400°C. During the experiment, the surface temperature is measured with a pyrometer with a small measurement point ($d = 1.5$ mm) and a high time resolution (1000 Hz). Therewith, it is possible to spatially and temporally resolve the spallation process [*Kant and Rudolf von Rohr*, 2016]. A uniaxial pressure p orthogonally to the front surface was applied to the rock cores during the spallation experiments with the confining system proposed by Hoeser et al. [*Hoeser*, 2016; *Meier et al.*, 2016]. In this system, the rock core is placed between two metal flanges. The mechanical load on the sample is applied with four screws pressing the flanges together. The resulting pressure is measured separately with a load cell, and the flame impinges on the rock through a circular hole in the flanges (see Figure 5).

Central Aare granite was selected as a suitable material for the validation. Spalling has been observed for this rock and due to the intensive research on this formation, the physical properties have been well documented in previous studies (see Table 1) [*Pahl et al.*, 1989; *Pilgerstorfer et al.*, 2013; *Rutqvist and Tsang*, 2004; *Keusen et al.*, 1989; *Kazerani and Zhao*, 2014; *Moeri et al.*, 2003]. The samples with a diameter of 85 mm and a length of 150 mm were taken from Chüenzentennlen close to the Grimsel Pass in Switzerland. The rock material has a classical granitic composition. The grain sizes are highly variable ranging from a few micrometer up to ~4 mm for feldspar, ~1 mm for plagioclase and biotite and ~0.9 mm for quartz. In Figure 6 an example of a thin section from the undamaged rock can be found.

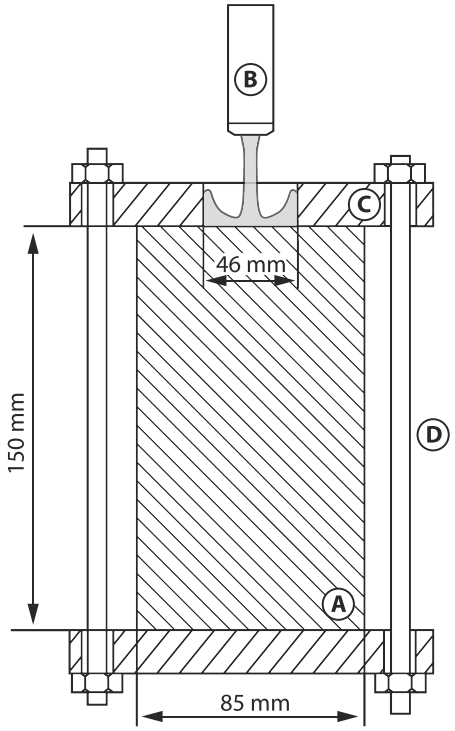


Figure 5. Illustration of the experimental setup used for the preliminary validation. (a) rock sample, (b) methane air burner, (c) metal flange, (d) M16 thread with nuts to confine the rock.

4.2. Theoretically Expected Values According to the Presented Model

The theoretically required temperature jump to initiate thermal spalling can be calculated with the proposed model, as shown in equation (15). Thereby, the temperature is assumed to be constant over the crack length.

$$\Theta = (T - T_i) = \frac{(1-\nu)}{E\alpha}\left[K_{IC}\sqrt{\frac{\pi}{4a}}\left(\int_0^1 f_K(\xi)\mathrm{d}\xi\right)^{-1} - p\frac{\nu}{(1-\nu)}\right] \quad (15)$$

For the calculation of the spalling temperature, the largest initially present cracks were selected, as they will have the lowest temperature required for thermal crack formation. Shorter cracks require higher surface temperatures before spalling appears and vice versa if higher temperatures are applied, more cracks can be activated and the spalling frequency will increase. The temperature difference was calculated with all possible combinations of the physical properties listed in Table 1 and summarized to a mean value with the according standard deviation (see Table 2). The presented model suggests that, without any additional confining pressure, spalling of the rock surface should appear at temperatures between 390°C and 560°C.

Table 1. Physical Properties of Central Aare Granite Used for Calculating the Maximal Possible Temperature Jump; Data Obtained From Literature; All Values Have Been Evaluated at Room Temperature

Physical Properties	Values	Ref.[a]
Young's modulus	$E = 30–40$ GPa	[1]
Poisson's ratio	$\nu = 0.19–0.33$	[2]
Fracture toughness	$K_{IC} = 1.5$ MPa m$^{0.5}$	[3]
Crack length	$a = 20$ µm	[4]
Coeff. of thermal exp.	$\alpha = (7.8–8.2) \cdot 10^{-6}$ 1/K	[5]
Initial temperature	$T_i = 20$°C	[-]

[a][1]: [*Pahl et al.*, 1989; *Pilgerstorfer et al.*, 2013; *Rutqvist and Tsang*, 2004], [2]: [*Pilgerstorfer et al.*, 2013; *Rutqvist and Tsang*, 2004; *Keusen et al.*, 1989], [3]: [*Kazerani and Zhao*, 2014], [4]: [*Moeri et al.*, 2003], [5]: [*Gens et al.*, 1998; *Rutqvist and Tsang*, 2004].

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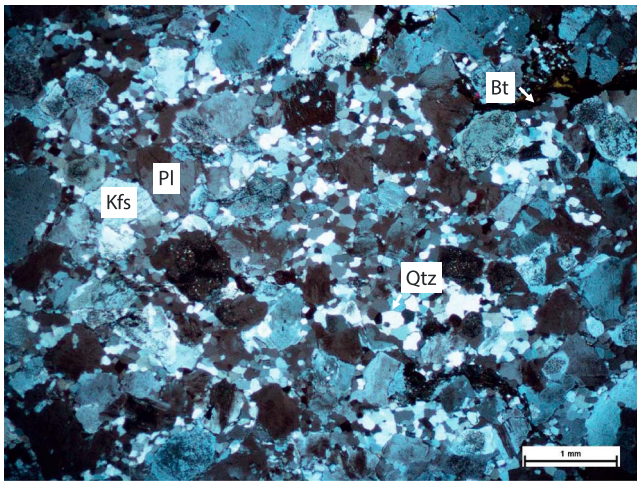


Figure 6. Thin section of Central Aare granite, used for the experimental validation of the presented spallability model. The rock is mainly composed of quartz (Qtz), K-feldspar (Kfs), plagioclase (Pl), and minor amount of biotite (Bt).

4.3. Experimental Results and Comparison With the Model

With the experimental setup introduced above, spallation experiments were conducted with three different external uniaxial confining pressures ($p = 0$ MPa, $p = 27$ MPa, and $p = 48$ MPa). As critical temperature for spalling, the temperature in the failure plane directly after the spall has detached from the surface was selected [*Kant and Rudolf von Rohr*, 2016; *Rauenzahn and Tester*, 1989; *Wilkinson and Tester*, 1993]. The evaluation algorithm proposed by *Kant and Rudolf von Rohr* [2016] was applied to the temperature raw data from the pyrometer.

If no uniaxial confining pressure is applied, temperatures in the crack plane of the rock between 553°C and 694°C were measured (see Figure 7). These values fit quite accurately to the theoretically derived values from the proposed spalling model which estimated, as discussed above, temperatures between 390°C and 560°C. With increasing uniaxial confining pressure, the measured spalling temperature decreases, as less additional thermal stresses are required to initiate spalling. This trend can be also observed in the theoretical values suggested by the proposed model. In all experiments, not the first initiated spall could be detected by the measurement method, as the spall did not detach in the localized measurement area of the pyrometer. After the first spall has detached from the surface, the surface temperature continues to rise. If higher temperatures are present, cracks with a shorter length can be initiated (see equation (15) and spalls will be detected also at this higher temperature level. Therefore, the experimental temperature values in Figure 7 are higher than the values predicted by the model, which only considers the onset of spallation, if the largest crack length is used for the calculation. Additionally, the values suggested by the model have been calculated with physical properties at room temperature. As physical properties of rocks vary significantly with increasing temperature, further inaccuracies are introduced to the evaluation. It can be concluded that the values calculated with the presented model are in reasonable agreement with the experimental values observed on Central Aare granite. Additionally, the selected crack geometry and the used assumptions seem to be able to describe thermal spalling of rocks in a simplified manner.

Table 2. Theoretically Expected Temperature Jump for Different Confining Pressures p

Confining Pressure p	θ Mean Value	θ Range
0 MPa	475°C	390–560°C
27 MPa	450°C	365–535°C
48 MPa	430°C	340–510°C

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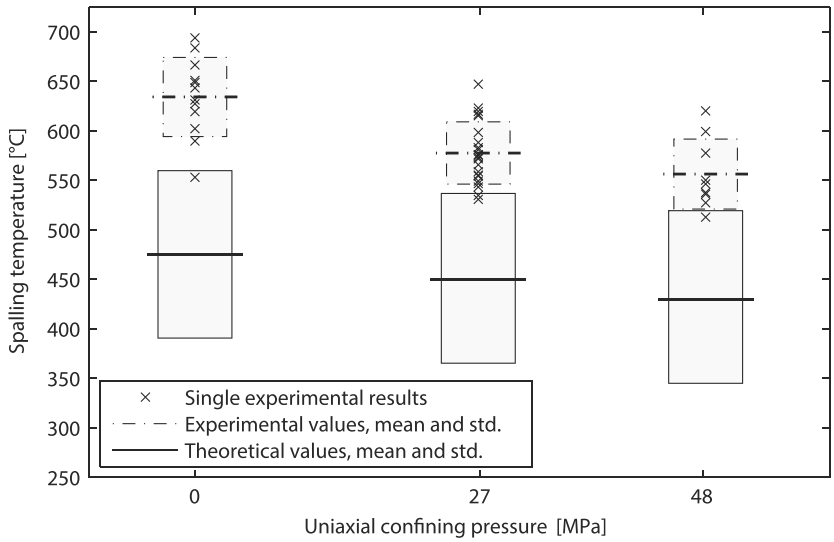


Figure 7. Temperature jump required to initiate spalling of Central Aare granite for different confining pressures p calculated with the presented model and experimental data of the spalling temperature shortly after the onset of spallation.

5. Conclusions

We present a model to estimate spalling and thermal cracking behaviors of rocks with a focus on thermal spallation drilling. The model is based on linear fracture mechanics and the stress intensity concept. It couples rapid heat transfer with induced thermal stresses and fracture-based mechanics in rocks.

With the model, the operating conditions required to initiate thermal rock spalling can be evaluated. Additionally, a Spallability number is presented allowing a straightforward application of the derived equations in various conditions and a classification of rocks according to their ability to spall. This number is divided into terms representing the physical properties of the rock, the heat transfer properties of the fluid, and the influence of confining pressures. The model follows already reported influences on the spallability of rocks. For thermal spalling the following rock properties are desired: low thermal conductivities [*Preston and White*, 1934; *Walsh and Lomov*, 2013], high Young's moduli [*Rauenzahn and Tester*, 1989; *Preston and White*, 1934; *Walsh et al.*, 2011], high vertical pressures [*Rauenzahn and Tester*, 1989; *Hoeser*, 2016], high thermal expansion coefficients [*Rauenzahn and Tester*, 1989; *Preston and White*, 1934], and low fracture toughness values [*Rauenzahn and Tester*, 1989; *Walsh et al.*, 2011].

A preliminary experimental validation of the model with a typical granitic rock was conducted. The model showed reasonable agreement with the experimental data. The measured temperature jump which is required to initiate surface spalling could be reproduced by the presented model.

Nomenclature

Roman symbols

a	Crack length (m)
E	Young's modulus (Pa)
Bi	Biot number (-)
h	Heat transfer coefficient (W/(m^2 K))
K	Stress intensity factor (Pa $\sqrt{m}$)
p	Pressure (Pa)
Sp	Spallability number (-)
t	Time (s)
T	Temperature (°C)
z	Distance from the surface (m)

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Greek letters

α	Linear thermal expansion coefficient (K^{-1})
ϵ	Strain component (-)
Θ	Temperature difference (K)
κ	Thermal diffusivity (m^2/s)
λ	Thermal conductivity (W/(m K))
ν	Poisson's ratio (-)
ξ	Dimensionless space (-)
σ	Stress vector (Pa)
τ	Dimensionless time (-)

Subscripts and superscripts

1	Orthogonally to the surface
I	Referring to mode I cracking
2	Parallel to the surface
fl	Referring to the fluid
i	Initial conditions
T	Referring to the time
K	Referring to the crack
xx	Normal to the *x* axis
yy	Normal to the *y* axis
zz	Normal to the *z* axis

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